

Three-Dimensional Evaluation Framework (D1/D2/D3)

D1: In-Domain NER

Mahānāma Test Split

6,672 sentences

Gold NER labels

Primary benchmark

Micro F1 + 95% CI

D2: Linguistic Capability

DCS Held-Out (200 sent.)

Tasks: S, L, LM, SLM

Preservation check

Final NER: S=85%, L=93%

Note: Seed inconsistency

D3: Cross-Domain

UD_Sanskrit-UFAL

230 sentences (Pañcatantra)

Binary (PROPN only)

Final NER: 73.4% recall

(+16 pp vs Best NER)

Evaluation Summary

D1 (Primary): All 15 models evaluated on identical 6,672-sentence test set

D2 (Linguistic): NER-only models score 0%; Final NER retains S=85%, L=93%

D3 (Cross-Domain): Binary PROPN detection only (no gold PER/LOC/MISC labels)

Statistical Tests: Bootstrap 95% CI (10,000 resamples) + McNemar's test

Key Finding: Final NER trades -0.8 F1 for +16 pp cross-domain recall + preserved D2

Limitation: D2 seed inconsistency across machines (seed=42 vs 123)